



STEM SMART Single Maths 33 - The Product Rule & Implicit Differentiation >

The Product and Quotient Rules

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level P3

Fill in the blanks below to complete the summary notes on the product and quotient rules.

The function $y = u(x)v(x)$ is a of the functions of $u(x)$ and $v(x)$. To differentiate a function like this we use the rule:

$$\frac{dy}{dx} = \text{} v + u \text{}$$

The function $y = \frac{u(x)}{v(x)}$ is a of the functions of $u(x)$ and $v(x)$. To differentiate a function like this we use the rule:

$$\frac{dy}{dx} = \frac{\text{} v - u \text{}}{v^2}$$

Items:

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Implicit Differentiation

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Differentiation Stage & difficulty: A Level P3

Fill in the blanks below to complete the summary notes on implicit differentiation.

Equations of the form $y = f(x)$, such as $y = 2x + 1$ or $y = 3 \tan x$, give y as an function of x .

Equations which are not in the form $y = f(x)$, such as $e^y + x^3 y^2 = 2x$, give y as an function of x .

To find the gradient of an implicit function, we need to know how to differentiate functions of y with respect to x .

e^y is a function of y . To differentiate a function of y with respect to x we use the rule.

$$\begin{aligned}\frac{d(e^y)}{dx} &= \frac{d(e^y)}{\text{d}y} \frac{\text{d}y}{dx} \\ &= \text{d}y \frac{dy}{dx}\end{aligned}$$

$x^3 y^2$ is a function of both x and y . To differentiate terms of this form with respect to x we first use the rule. Then, to differentiate the function of y , we use the chain rule.

$$\begin{aligned}\frac{d(x^3 y^2)}{dx} &= \frac{d(x^3)}{dx} y^2 + x^3 \frac{d(y^2)}{dx} \\ &= \frac{d(x^3)}{dx} y^2 + x^3 \frac{d(y^2)}{dy} \frac{dy}{dx} \\ &= \text{d}x + \text{d}y \frac{dy}{dx}\end{aligned}$$

Items:

chain explicit implicit product dx dy $3x^2 y^2$ e^y $2x^3 y$

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Differentiation: Products 2ii

Subject & topics: Maths Stage & difficulty: A Level P2

Given that $y = 4x^2 \ln x$, answer the following.

Part A

First Derivative

Find an expression for $\frac{dy}{dx}$.

The following symbols may be useful: `Derivative(y, x)`, `ln()`, `log()`, `x`, `y`

Part B

Second Derivative

Find the value of $\frac{d^2y}{dx^2}$, when $x = e^2$.

The following symbols may be useful: `Derivative(y , x , x)`, `ln()`, `log()`, `x`, `y`

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Differentiation: Products 4ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Differentiate with respect to x , simplifying your answers where possible.

Part A

$$\sin x \tan x$$

Differentiate $\sin x \tan x$.

The following symbols may be useful: x

Part B

$$x^2(x+1)^6$$

Differentiate $x^2(x+1)^6$.

The following symbols may be useful: x

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Differentiation: Quotients 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Differentiate with respect to x , simplifying your answers where possible.

Part A

$$\frac{\ln x}{x}$$

$$y = \frac{\ln x}{x}$$

The following symbols may be useful: `Derivative(y, x)`, `ln()`, `log()`, `x`, `y`

Part B

$$\frac{x^2}{\ln x}$$

$$y = \frac{x^2}{\ln x}$$

The following symbols may be useful: `Derivative(y, x)`, `ln()`, `log()`, `x`, `y`

Part C

Stationary point of $y = \frac{x^2}{\ln x}$

Determine the exact x -coordinate of the stationary point of the curve $y = \frac{x^2}{\ln x}$.

The following symbols may be useful: `e`, `ln()`, `x`

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Differentiation: Quotients 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A curve has equation $y = \frac{x^2+4}{x+2}$.

Part A

Derivative

Find an expression for $\frac{dy}{dx}$ in terms of x .

The following symbols may be useful: `Derivative(y, x)`, `x`, `y`

Part B

Normal

Find the equation of the normal to the curve at the point $(1, \frac{5}{3})$, giving your answer in the form $ax + by + c = 0$, where a , b , and c are integers.

The following symbols may be useful: `x`, `y`

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Implicit Differentiation 1

Pre-Uni Maths for Sciences J6.7

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level P3

Part A

Find $\frac{dy}{dx}$ if $x^2 + y^2 = r^2$.

Find $\frac{dy}{dx}$ if $x^2 + y^2 = r^2$, giving your answer as a simple function of x and y .

The following symbols may be useful: `Derivative(, x)`, `x`, `y`

Part B

Find gradient of tangent to $x^2 - xy + y^2 = 7$

Consider the curve $x^2 - xy + y^2 = 7$.

Find as a function of x and y the gradient of the tangent to the curve $x^2 - xy + y^2 = 7$.

The following symbols may be useful: `Derivative(, x)`, `x`, `y`

Hence evaluate the slope at the point $(-1, 2)$.

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Differentiation: Products 4i

Subject & topics: Maths **Stage & difficulty:** A Level P2

The equation of a curve has the form $y = e^{x^2}(ax^2 + b)$, where a and b are non-zero constants.

Part A

First Derivative

Find an expression for $\frac{dy}{dx}$.

The following symbols may be useful: `Derivative(y, x)`, `a`, `b`, `e`, `ln()`, `log()`, `x`, `y`

Part B

Second Derivative

Find an expression for $\frac{d^2y}{dx^2}$.

The following symbols may be useful: `Derivative(y, x, x)`, `a`, `b`, `e`, `ln()`, `log()`, `x`, `y`

Part C **a in terms of b**

It is given that $\frac{d^2y}{dx^2}$ can be expressed in the form $e^{x^2}(cx^4 + d)$, where c and d are non-zero constants. Find an expression for a in terms of b .

The following symbols may be useful: a , b

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Differentiation: Quotients 2i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Derivative

Given that $y = \frac{4\ln(x)-3}{4\ln(x)+3}$, find an expression for $\frac{dy}{dx}$.

The following symbols may be useful: `Derivative(y, x)`, `ln()`, `log()`, `x`, `y`

Part B

Gradient

Give the exact value of the gradient of the curve $y = \frac{4\ln(x)-3}{4\ln(x)+3}$ at the point where it crosses the x -axis.

The following symbols may be useful: `Derivative(y, x)`, `e`

Part C

Area

Figure 1 shows part of the curve with equation

$$y = \frac{2}{x^{\frac{1}{2}}(4\ln(x) + 3)}.$$

The region shaded in the diagram is bounded by the curve and the lines $x = 1$, $x = e$, and $y = 0$. Find the exact value of the integral I where

$$I = \int_1^e \pi y^2 dx.$$

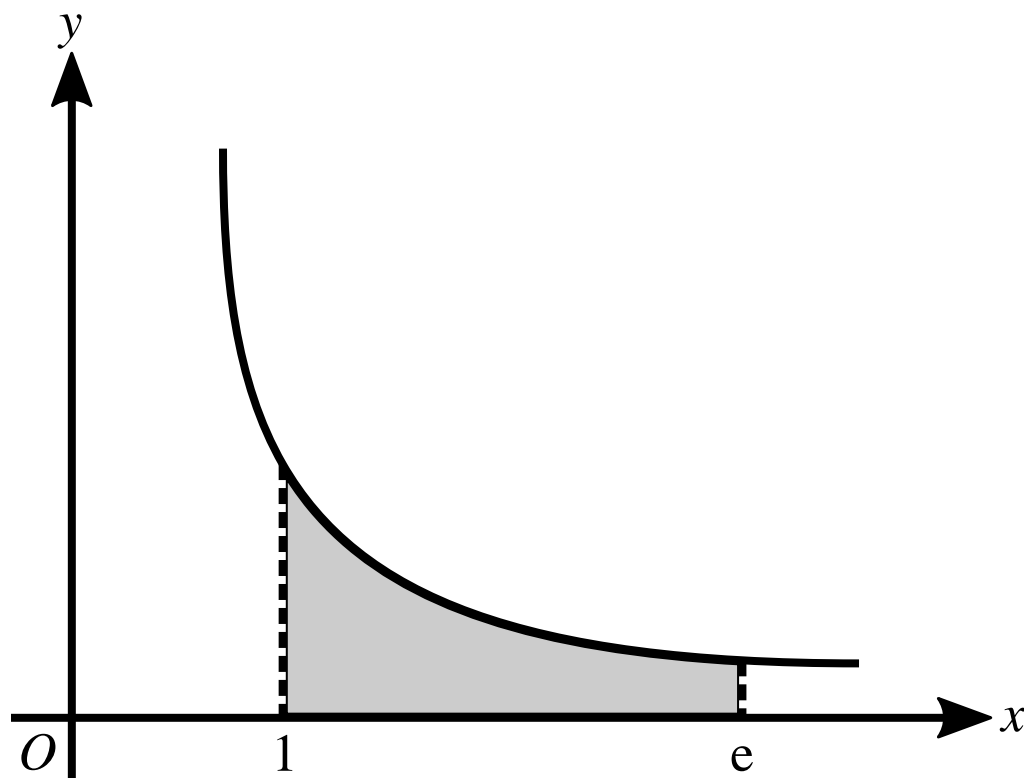


Figure 1: A diagram showing part of the curve with equation $y = \frac{2}{x^{\frac{1}{2}}(4\ln(x)+3)}$.

Give the exact value of I .

The following symbols may be useful: I , π

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Calculus: Inverse Trigonometry 2i

Subject & topics: Maths

Stage & difficulty: Further A P2

Part A

Derivative of $\arcsin x$

Find the derivative of $\arcsin x$

The following symbols may be useful: x

Part B

Implicit differentiation

Given that

$$\arcsin 2x + \arcsin y = \frac{1}{2}\pi$$

find the exact value of $\frac{dy}{dx}$ when $x = \frac{1}{4}$.

Adapted with permission from UCLES, A Level, January 2009, Paper 4726, Question 3.

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